PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Programming Fundamentals Lab 09	**Instructor: Mr. Muhammad Hamza**
Program: AI Fall 2025	**Date: 21th December, 2025**
Marks: 10	**Lab Engineer: Qandil Fatima**
Student Name: Eshal Ayub	**Reg. No: B25F1061AI100**

Instructions:

- Submit the file with your names following your registration numbers like **AI001_Name.**
- **Copy Paste work will be given no marks.**

Lab Task

- **Task1:**

Write C++ Program to Store Information of a Student in a Structure. It stores the information (name, roll and marks entered by the user) of a student in a structure and displays it on the screen.

```cpp
#include <iostream>
#include <string>
using namespace std;

struct Student {
    string name;
    int roll;
    float marks;
};

int main() {
    Student s;

    cout << "Enter name: ";
    getline(cin, s.name);

    cout << "Enter roll number: ";
    cin >> s.roll;

    cout << "Enter marks: ";
    cin >> s.marks;

    cout << "\nDisplaying Information:\n";
    cout << "Name: " << s.name << endl;
    cout << "Roll: " << s.roll << endl;
    cout << "Marks: " << s.marks << endl;

    return 0;
}
```

Output:

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled4\cmake-build-debug\untitled4.exe"
Enter name: Eshal Ayub

Enter roll number:B25F1061AI100

Enter marks:
Displaying Information:
Name:  Eshal Ayub
Roll: 0
Marks: 6.50202e-43

Process finished with exit code 0
```

- **Task2:**

Write a C++ program to calculate difference between two time period.

```cpp
#include <iostream>
using namespace std;
struct Time {
    int hours;
    int minutes;
    int seconds;
};
int main() {
    Time t1, t2, diff;
    cout << "Enter first time (h m s): ";
    cin >> t1.hours >> t1.minutes >> t1.seconds;
    cout << "Enter second time (h m s): ";
    cin >> t2.hours >> t2.minutes >> t2.seconds;
    // Convert both to seconds
    int total1 = t1.hours * 3600 + t1.minutes * 60 + t1.seconds;
    int total2 = t2.hours * 3600 + t2.minutes * 60 + t2.seconds;
    int d = total1 - total2;
    if (d < 0) d = -d;
    diff.hours   = d / 3600;
    d            = d % 3600;
    diff.minutes = d / 60;
    diff.seconds = d % 60;
    cout << "Difference: "
         << diff.hours << "h "
         << diff.minutes << "m "
         << diff.seconds << "s\n";

    return 0;
}
```

Output:

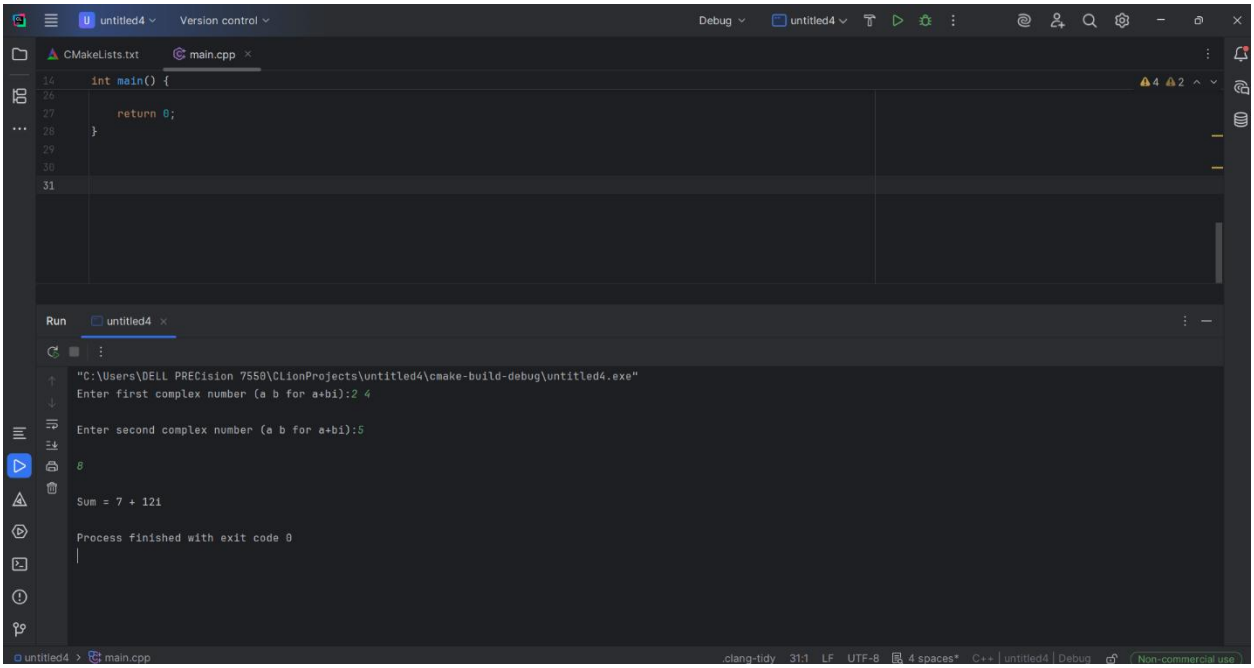
```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled4\cmake-build-debug\untitled4.exe"
Enter first time (h m s):11 33 52

Enter second time (h m s):8 12 15

Difference: 3h 21m 37s

Process finished with exit code 0
```

- **Task3:**

Write a C++ Program to Add Complex Numbers by Passing Structure to a Function. Where a complex number can be represented as a+bi .

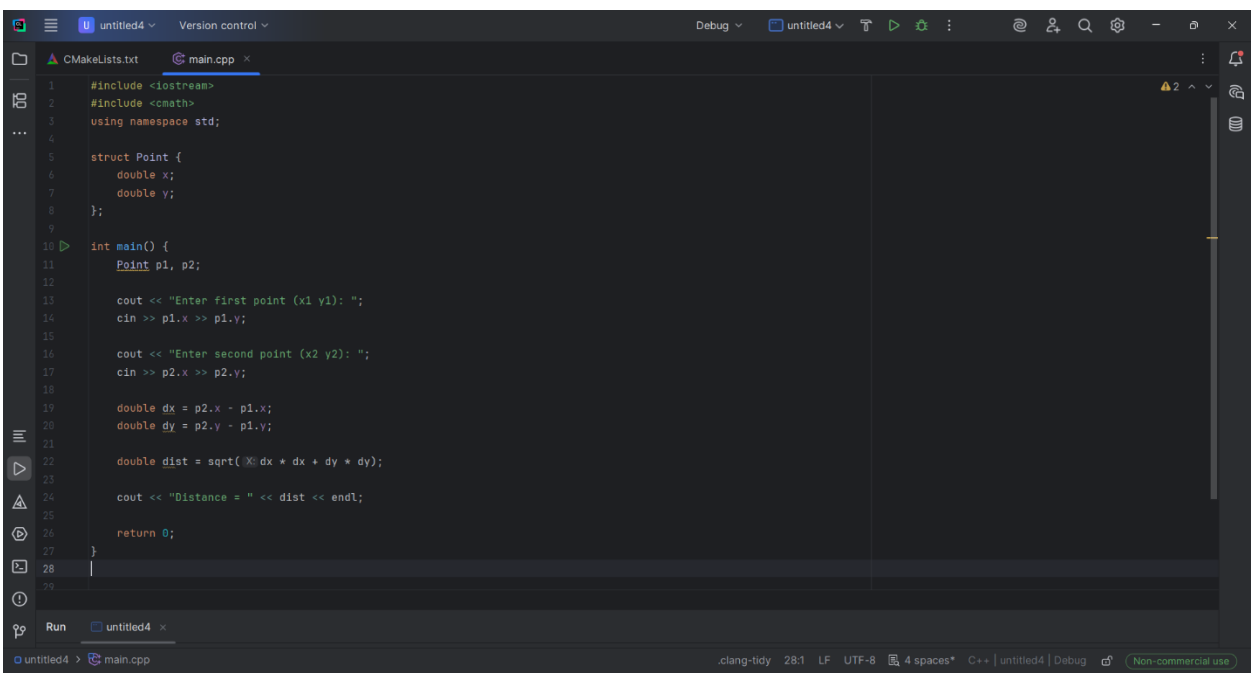
```cpp
#include <iostream>
using namespace std;

struct Complex {
    float real;
    float imag;
};
Complex addComplex(Complex c1, Complex c2) {
    Complex result;
    result.real = c1.real + c2.real;
    result.imag = c1.imag + c2.imag;
    return result;
}
int main() {
    Complex c1, c2, sum;

    cout << "Enter first complex number (a b for a+bi): ";
    cin >> c1.real >> c1.imag;

    cout << "Enter second complex number (a b for a+bi): ";
    cin >> c2.real >> c2.imag;

    sum = addComplex(c1, c2);

    cout << "Sum = " << sum.real << " + " << sum.imag << "i\n";

    return 0;
}
```

Output:



- **Task4:**

Write a C++ program to calculate distance between two points (on Cartesian plane) using Euclidean distance i.e. sqrt((x2-x1)^2+(y2-y1)^2).



Output:

```cpp
int main() {

    double dist = sqrt( dx * dx + dy * dy);

    cout << "Distance = " << dist << endl;

    return 0;
}
```

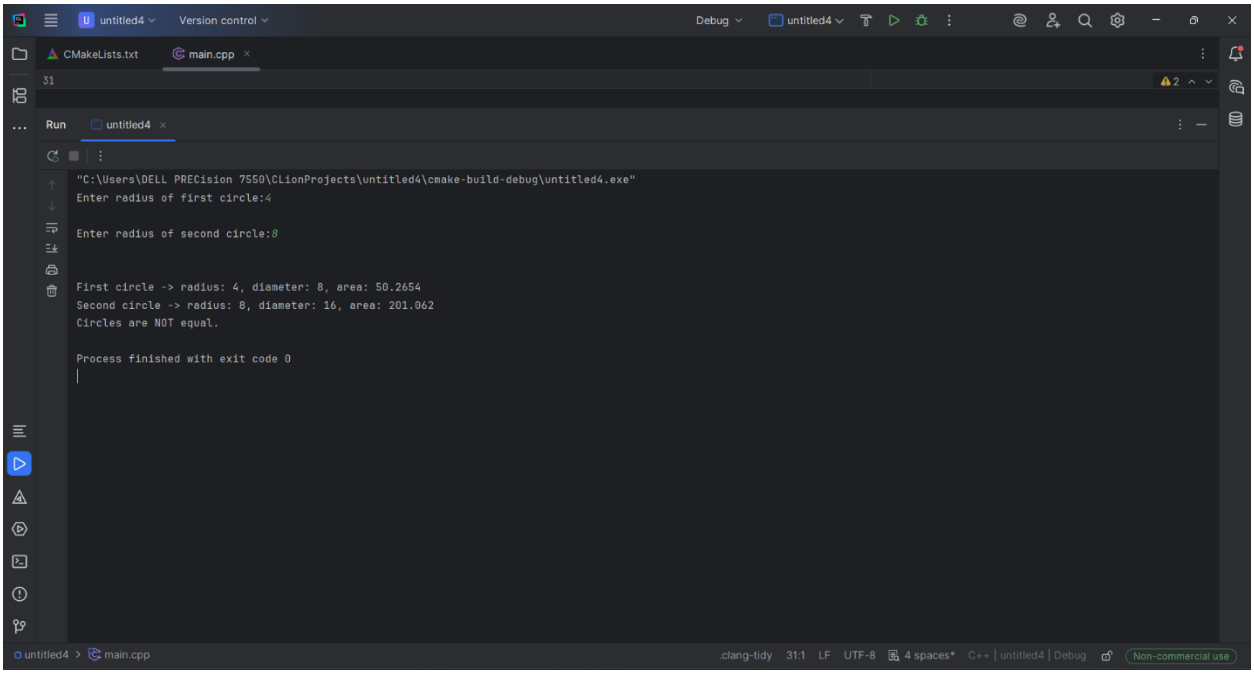
```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled4\cmake-build-debug\untitled4.exe"
Enter first point (x1 y1):4 8

Enter second point (x2 y2):2 6

Distance = 2.82843

Process finished with exit code 0
```

- **Task 5:**

Write a program in C++ to stores data about a circle (radius, diameter & diameter) in a structure. Create two circle objects and compare if the two circles are equal or not.

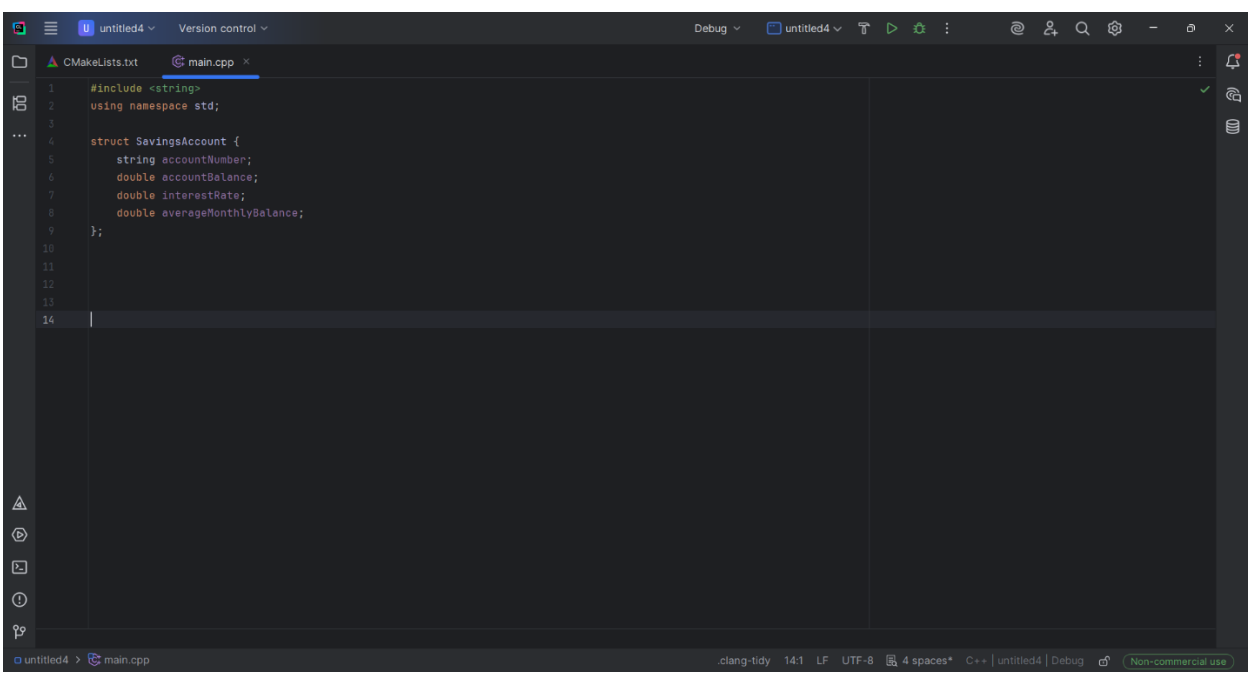
```cpp
#include <iostream>
using namespace std;
struct Circle {
    double radius;
    double diameter;
    double area;
};
int main() {
    Circle c1, c2;
    cout << "Enter radius of first circle: ";
    cin >> c1.radius;
    c1.diameter = 2 * c1.radius;
    c1.area     = 3.14159 * c1.radius * c1.radius;
    cout << "Enter radius of second circle: ";
    cin >> c2.radius;
    c2.diameter = 2 * c2.radius;
    c2.area     = 3.14159 * c2.radius * c2.radius;
    cout << "\nFirst circle -> radius: " << c1.radius
         << ", diameter: " << c1.diameter
         << ", area: " << c1.area << endl;
    cout << "Second circle -> radius: " << c2.radius
         << ", diameter: " << c2.diameter
         << ", area: " << c2.area << endl;
    if (c1.radius == c2.radius) {
        cout << "Both circles are equal.\n";
    } else {
        cout << "Circles are NOT equal.\n";
    }
    return 0;
}
```

Output:



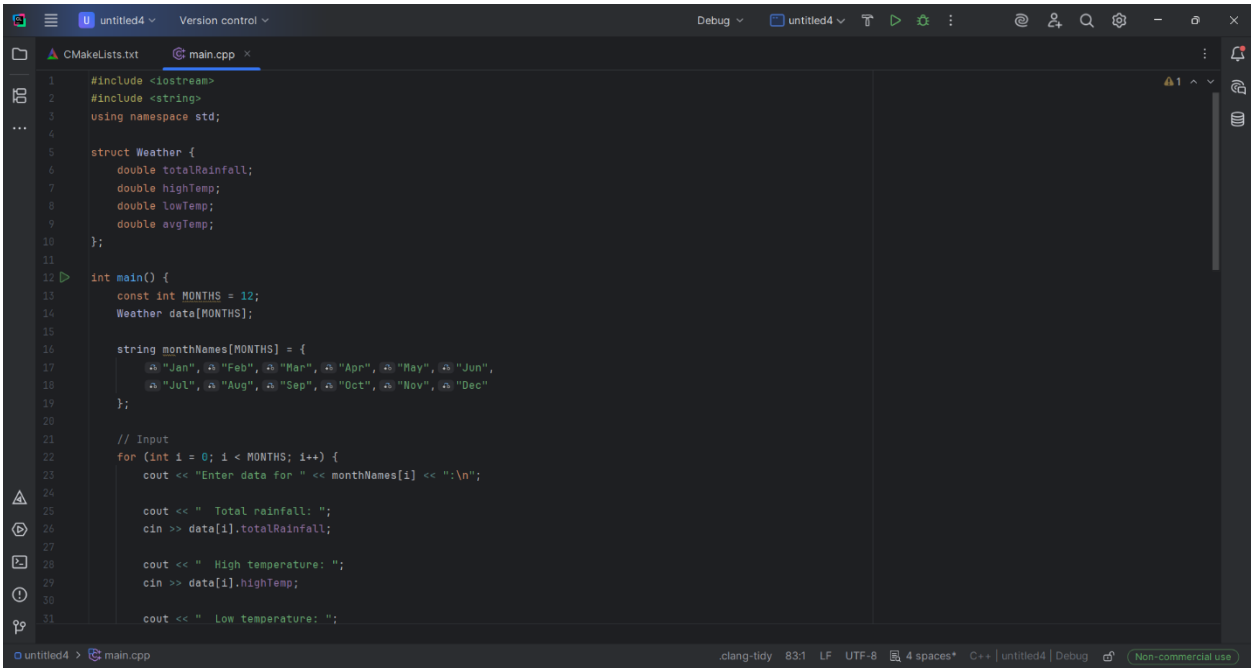
- **Task 6:**

Write a structure declaration to hold the following data about a savings account: Account Number (string object) Account Balance (double) Interest Rate (double) Average Monthly Balance (double)



- **Task 7:**

Write a program that uses a structure to store the following weather data for a particular month: Total Rainfall High Temperature Low Temperature Average Temperature The program should have an array of 12 structures to hold weather data for an entire year. When the program runs, it should ask the user to enter data for each month. (The average temperature should be calculated.) Once the data are entered for all the months, the program should calculate and display the average monthly rainfall, the total rainfall for the year, the highest and lowest temperatures for the year (and the months they occurred in), and the average of all the monthly average temperatures.



```cpp
#include <iostream>
#include <string>
using namespace std;

struct Weather {
    double totalRainfall;
    double highTemp;
    double lowTemp;
    double avgTemp;
};

int main() {
    const int MONTHS = 12;
    Weather data[MONTHS];

    string monthNames[MONTHS] = {
        "Jan", "Feb", "Mar", "Apr", "May", "Jun",
        "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"
    };

    // Input
    for (int i = 0; i < MONTHS; i++) {
        cout << "Enter data for " << monthNames[i] << ":\n";

        cout << "  Total rainfall: ";
        cin >> data[i].totalRainfall;

        cout << "  High temperature: ";
        cin >> data[i].highTemp;

        cout << "  Low temperature: ";
```

```cpp
int main() {
    for (int i = 0; i < MONTHS; i++) {

        cout << "  Low temperature: ";
        cin >> data[i].lowTemp;

        data[i].avgTemp = (data[i].highTemp + data[i].lowTemp) / 2.0;

        cout << endl;
    }

    // Calculations
    double totalRainYear = 0;
    double avgMonthlyRain = 0;
    double sumAvgTemps = 0;

    double highestTemp = data[0].highTemp;
    int highMonthIndex = 0;

    double lowestTemp = data[0].lowTemp;
    int lowMonthIndex = 0;

    for (int i = 0; i < MONTHS; i++) {
        totalRainYear += data[i].totalRainfall;
        sumAvgTemps += data[i].avgTemp;

        if (data[i].highTemp > highestTemp) {
            highestTemp = data[i].highTemp;
            highMonthIndex = i;
        }
```

```cpp
int main() {
    for (int i = 0; i < MONTHS; i++) {
            highestTemp = data[i].highTemp;
            highMonthIndex = i;
        }

        if (data[i].lowTemp < lowestTemp) {
            lowestTemp = data[i].lowTemp;
            lowMonthIndex = i;
        }
    }

    avgMonthlyRain = totalRainYear / MONTHS;
    double avgOfMonthlyAverages = sumAvgTemps / MONTHS;

    // Output
    cout << "\nAverage monthly rainfall: " << avgMonthlyRain << endl;
    cout << "Total rainfall for the year: " << totalRainYear << endl;

    cout << "Highest temperature: " << highestTemp
         << " in " << monthNames[highMonthIndex] << endl;

    cout << "Lowest temperature: " << lowestTemp
         << " in " << monthNames[lowMonthIndex] << endl;

    cout << "Average of all monthly average temperatures: "
         << avgOfMonthlyAverages << endl;

    return 0;
}
```

Output:

```
"C:\Users\DELL PRECision 7550\CLionProjects\untitled4\cmake-build-debug\untitled4.exe"
Enter data for Jan:
  Total rainfall:5

  High temperature:9

  Low temperature:0


Enter data for Feb:
  Total rainfall:4

  High temperature:1

  Low temperature:-5


Enter data for Mar:
  Total rainfall:7

  High temperature:6

  Low temperature:1


Enter data for Apr:
  Total rainfall:9
```

```
Enter data for Apr:
  Total rainfall:9

  High temperature:7

  Low temperature:3


Enter data for May:
  Total rainfall:6

  High temperature:20

  Low temperature:14


Enter data for Jun:
  Total rainfall:0

  High temperature:35

  Low temperature:23


Enter data for Jul:
  Total rainfall:1

  High temperature:45
```

```
  High temperature:45

  Low temperature:30


Enter data for Aug:
  Total rainfall:2

  High temperature:40

  Low temperature:35


Enter data for Sep:
  Total rainfall:4

  High temperature:

41

  Low temperature:37


Enter data for Oct:
  Total rainfall:3

  High temperature:30
```

```
  Low temperature:25


Enter data for Nov:
  Total rainfall:29

  High temperature:30

  Low temperature:25


Enter data for Dec:
  Total rainfall:0

  High temperature:20

  Low temperature:3



Average monthly rainfall: 5.83333
Total rainfall for the year: 70
Highest temperature: 45 in Jul
Lowest temperature: -5 in Feb
Average of all monthly average temperatures: 19.7917

Process finished with exit code 0
```